First Tool-Name Loss Weight: Three Instead of One

Status: Owner training and evaluation completed; paired result review completed. Date: 2026-10-15.

Hypothesis

The frozen name-first model still makes wrong routing choices on selected seen inputs. Test whether directly emphasizing its first call name improves routing, without changing the data mixture, prompt, serialization, or epoch count.

The preceding component diagnostic does not prove this intervention is best. Question/reason loss is larger, but exact-target loss is not semantic adequacy. This experiment deliberately targets routing rather than text reproduction.

Fixed Reference

Use the fresh two-epoch QKVO name-first recipe, not A's older Q/V-only recipe:

- base: `Qwen/Qwen2.5-1.5B-Instruct`, initialized from pretrained weights;
- fresh LoRA: rank 8, alpha 16, Q/K/V/O, dropout 0.05;
- learning rate `2e-4`, two epochs, seed 1, same scheduler and warmup;
- unchanged version-14 name-first split files and order;
- raw caps: 1,000 train and 200 validation;
- laptop overrides: context 1,024, batch 1, gradient accumulation 16.

Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. Reference config hash: `684a76cd39d845b6177bdaee495bb77bcb436406d384baf912f80b8164fb8b8b`. Weighted config hash: `7a5c074667c6041c6adc47e890197c5ce07f8a1175e90f4416f65592c027f619`.

One Objective Change

`training.first_tool_name_weight = 3.0` applies to pure string-content tokens of the first tool name in every class: ACT, CLARIFY and REFUSE. Other assistant tokens, including questions, reasons, arguments and later call names, keep weight 1. Mixed text/syntax tokens also stay at 1, matching the diagnostic's region policy. Prompt and padding labels stay ignored. No examples are repeated or rewritten.

Use the same shared character-offset mapping as the diagnostic. The training token IDs, labels, retained rows and truncation behavior are unchanged. Validation has no extra weights and retains ordinary assistant-token cross-entropy.

Normalize training loss as `sum(weight * token_nll) / sum(weights)` over the actual accumulation window, including a shorter final window. Three is a relative weight, not a blanket threefold increase in loss or learning rate. Gradient tests check equivalence to the corresponding combined weighted batch. The custom path supports one training device and fails explicitly on incompatible Trainer APIs.

Historical configs default to weight 1 and use the original Trainer. Their config hashes are unchanged. Nondefault weighting is included in the new config hash.

Command

From `project-baymax`:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first_
tool_weight3.json"

.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

This does not resume the existing adapter. The new config names its own model folder, ending in `_qkvo_name_first_tool_weight3`, and refuses a populated output directory before loading ML dependencies. Existing checkpoints remain untouched.

The new model directory receives `weighted-loss-run.json` before training, with the full config, concrete plan, retained row count, weighted first-name-token count, effective token mass, and normalization/validation definitions.

Evaluation Plan

Use exactly the same 150 established development scenarios and compact JSON inference presentation as the unweighted name-first reference. Preserve the parser, scorer, decoding settings and all old results. Compare routing macro-F1, the full decision matrix, task success and invalid-output rate; inspect wording adequacy separately. The repeatedly inspected suite is not an untouched final test.

Training loss is now weighted while validation loss is not. Do not compare the two values as if they use the same objective, or compare the weighted training loss directly against the historical unweighted training loss.

Measured Outcome

The owner completed training and the compact-JSON evaluation. The reported usable artifact contains 974 training rows and 195 validation rows, with no exact duplicates. Training retains 400 BAYMAX rows, including 100 clarification and 100 refusal rows. The saved `weighted-loss-run.json` records 3,275 weighted first-name tokens and effective training token mass 43,164.

Metric	Unweighted name-first	First-name weight 3
Task successes / 150	57	46
Routing macro-F1	0.612	0.626
Correct routing decisions / 150	99	99
Invalid outputs / 150	1	3
Average argument accuracy	0.835	0.787

Weighting increased correct clarification decisions from 10 to 15, but correct ACT decisions fell from 66 to 64 and correct refusals from 23 to 20. It is not an overall task-performance win. The conditional decision-accuracy increase excludes more unknown decisions; it is not an increase in total correct routes.

The paired failure review documents all 15 lost successes and four gained successes, separating real execution/plan regressions from wording-score changes. Keep unweighted name-first as this experiment's reference; do not change evaluation rules to promote either model.

The coding agent inspected saved results and run metadata only. It did not run training, inference, a benchmark comparison script, or rescoring. Local experiment Markdown remains uncommitted under the owner's documentation restriction.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.